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Spurious tones in digital delta-sigma modulators resulting from pseudorandom dither

Michael Peter Kennedy[a,b,*], Hongjia Mo[a,b], Brian Fitzgibbon[c]

[a]School of Engineering—Electrical and Electronic Engineering, University College Cork, Cork, Ireland
[b]Microelectronic Circuits Centre Ireland, Tyndall National Institute, Cork, Ireland
[c]Susquehanna International Group, Dublin, Ireland

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Abstract

Digital delta-sigma modulators (DDSMs) are finite state machines; their spectra are characterized by strong periodic tones (so-called spurs) when they cycle repeatedly in time through a small number of states. This happens when the input is constant or periodic. Pseudorandom dither generators are widely used to break up periodic cycles in DDSMs in order to eliminate spurs produced by underlying periodic behavior. Unfortunately, pseudorandom dither signals are themselves periodic and therefore can have limited effectiveness. This paper addresses the fundamental limitations of using pseudorandom dither signals that are inherently periodic. We clarify some common misunderstandings in the DDSM literature. We present rigorous mathematical analysis, case studies to illustrate the issues, and insights which can prove useful in design.

1. Introduction

A digital delta-sigma modulator (DDSM) is commonly used to control the multimodulus divider in the feedback path of a fractional-N frequency synthesizer. The DDSM allows the synthesizer to achieve high frequency resolution. However, it also introduces quantization noise

*Corresponding author at: School of Engineering—Electrical and Electronic Engineering, University College Cork, Cork, Ireland.

E-mail addresses: peter.kennedy@ucc.ie (M.P. Kennedy), hongjia.mo@tyndall.ie (H. Mo), b.fitzgibbon@umail.ucc.ie (B. Fitzgibbon).

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which manifests itself as phase noise at the output of the synthesizer. The quantization noise introduced by the DDSM is highpass-filtered by the noise transfer function (NTF) of the DDSM. This shaped quantization noise is integrated once and then lowpass filtered by the transfer function of the synthesizer, before appearing as phase noise at the output [1]. If the classical model of quantization (CMQ [2]) applies, it is possible in principle to hide the shaped quantization noise below the phase noise envelope associated with the intrinsic and extrinsic noise sources in the underlying synthesizer [1]. In practice, however, the DDSM may introduce unwanted *periodic* components at the output; these are called "spurious tones" (spurs). A DDSM is a nonlinear finite state machine with a deterministic dynamic; consequently, it inherently produces a periodic output when its input is constant or periodic.

In the frequency synthesizer application, the input to the DDSM is typically held constant in order to produce a local oscillator (LO) signal at a well-defined channel frequency. When the input to the DDSM is constant, its output is periodic. A periodic signal with period L_s is characterized by a frequency spectrum which contain components at DC and integer multiples of the fundamental frequency f_s/L_s, where f_s is the frequency at which the DDSM is clocked. If L_s is small, the energy of the quantization noise is concentrated in a small number of tones; these unwanted spectral components produce spurs in the phase noise spectrum of the synthesizer.

Over the past decade, much research has been carried out to find ways to reduce the magnitudes of spurs introduced by DDSMs. Two classes of techniques have been developed: (a) deterministic, which attempt to spread the energy of the quantization noise signal over as many tones as possible by maximizing the period L_s, and (b) stochastic, which break up the inherent periodicity of the signal by adding a random dither signal. State of the art deterministic techniques include setting predefined inputs and/or initial conditions [3–5], using prime modulus quantizers [6], and architectural modifications [7–9]. The most popular *stochastic* technique is the use of additive least significant bit (LSB) dithering [10,11] which can be introduced at various points of the DDSM architecture to break up periodic cycles [12]. In principle, the random additive LSB dither signal is non-periodic and the resulting phase noise spectrum of the DDSM is smooth. In practice, however, the stochastic technique uses a dither signal which is not random. Rather, it is usually produced by another finite state machine [13,14]. The most common approach is to use as dither the signal produced by a pseudorandom binary sequence (PRBS) generator; note that this is a *periodic* signal.

In this work, we ask what happens when the input to the DDSM is constant and the dither signal is periodic. There are some conflicting results in the literature on this subject. One manufacturer's datasheet claims that dithering increases the sequence length to that of the PRBS, while a recent paper suggests that an 8-bit LFSR dither generator is especially "efficient" at eliminating spurs [12]. We present mathematically rigorous results concerning the periodicity of the quantization noise produced by a DDSM having a constant input and a periodic dither signal. We explain clearly why dither signals with short periods may not be effective in removing spurs from the underlying DDSM. To illustrate these issues, we focus on a MASH architecture which incorporates a dither strategy using an 8-bit LFSR dither generator [12]. Although we concentrate on a popular MASH structure, the methods we use can be applied to any DDSM architecture.

This paper is organized as follows. In Section 2, we motivate the work by presenting spectra for a state-of-the-art dithered MASH DDSM architecture [12]. We highlight the differences between the predicted and simulated spectra with random and pseudorandom additive LSB dither, illustrating how we will predict the worst case performance.

We describe in Section 3 a simplified model of the MASH 1–1–1 DDSM with (filtered) additive LSB dither. We illustrate the dependence of the period of the output signal on the input and the initial state of the DDSM, as well as the initial state of the LFSR, in the case of zeroth- and first-order filtered additive LSB dither in Sections 4–6.

Specifically, we show how to calculate the period of the output of a MASH DDSM with a power-of-two modulus M and a periodic LFSR dither input with period L_{sd}. We demonstrate that the period can be as small as $4 \times L_{sd}$ and as large as $2M \times L_{sd}$, where M is the modulus of the quantizer in the DDSM, depending on the initial conditions of the modulator and the dither generator. Furthermore, we show that the observed spectrum of the output may contain two distinct sets of filtered periodic components that originate respectively in (a) the DDSM itself and (b) the dither generator. We show how to calculate the heights of the spurs due to the periodic dither.

We discuss our observations and summarize our conclusions in Section 8.

2. Motivation

Throughout this paper, we study the MASH 1–1–1 DDSM shown in Fig. 1; this comprises three cascaded first order error feedback modulators (EFM1s) and an error cancelation network [15]. Filtered additive LSB dither is modeled by the one-bit dither signal $d[n]$ and the discrete-time filter with transfer function $V(z)$.

Consider first the undithered case ($d[n]=0$). If the quantization noise introduced by the DDSM itself is white then the shaped quantization noise that appears at its output should be bounded by an envelope which rises at +60 dB/decade, as shown by the solid black curve (labeled “Theoretical noise figure”) in Fig. 2. When the input is constant, the DDSM may produce a periodic output with a short period. In particular, when the input is half-scale, the signal has a period $L_s=4$. In the frequency domain, this signal is characterized by four tones, as indicated by the crosses in Fig. 2. In this example, $f_s=25$ MHz, so the fundamental is at 6.25 MHz.

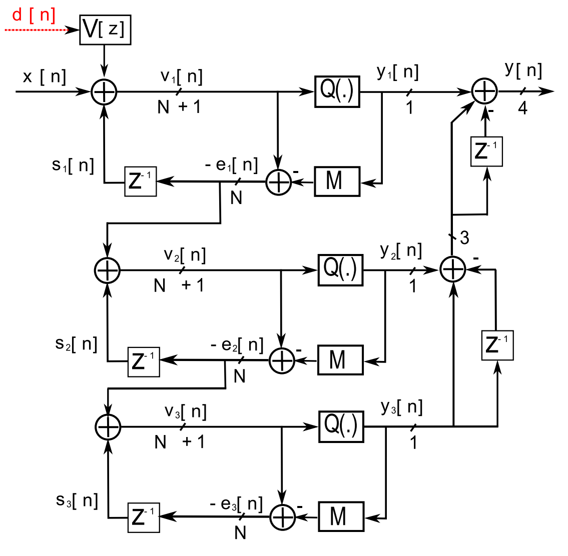

Fig. 1. Block diagram of a MASH 1–1–1 DDSM (DDSM3). It comprises three EFM1 stages with outputs y_1, y_2, and y_3, respectively, and an error cancelation network which combines these signals to produce the output y. The dither signal d is filtered by the transfer function $V(z)$ before being added to the input x.

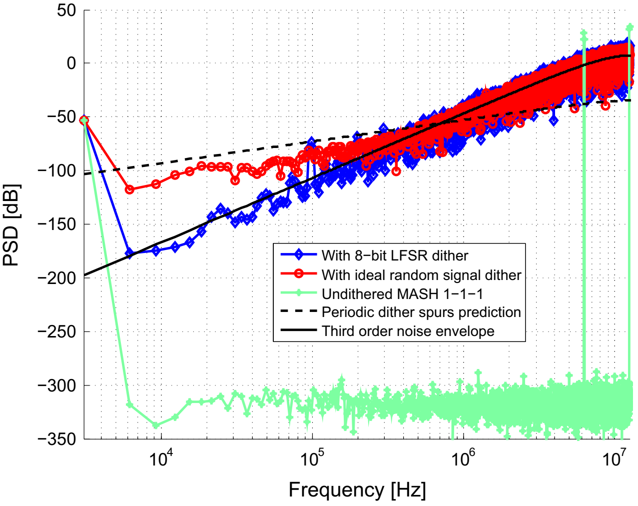

Fig. 2. Output spectra of MASH 1–1–1 (after [12]). The solid black curve shows the expected envelope of the third order shaped quantization noise, assuming that the CMQ applies. The green crosses show a worst-case spectrum with a half-scale input and no dither; this is characterized by a small number of strong tones. With first-order shaped random dither d, one obtains the red circles. If d is produced by an 8-bit LFSR, the spectrum is as shown by the blue diamonds. The dashed curve, the formula for which will be derived in this paper, indicates the upper bound on the frequency components due to dither. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

Filtered dither can be used to break the periodicity of the underlying DDSM. Fig. 3 shows a proposed architecture for accomplishing this [12].

By adding LSB dither to the second and third stages of a MASH 1–1–1, the spectrum appears to approach the ideal, as shown by the blue diamonds in Fig. 2. Somewhat counterintuitively, a periodic 8-bit LFSR dither signal appears to be more effective than a truly random signal (red circles) at reducing the low-frequency quantization noise.

In the following sections, we will explain mathematically the effects that are manifest in Fig. 2. The “random” signal appears as additive first-order shaped dither in the output spectrum, as expected. By contrast, we will show that the contribution of the 8-bit LFSR appears in the output as a *periodic* signal. Furthermore, we will derive an explicit expression to predict the heights of the largest spectral components (the dashed curve) in the dithered case. While the 8-bit case may appear to lie closer to the ideal (solid) third-order shaped envelope, it contains spurs which reach the dashed curve in Fig. 2.

3. MASH DDSM model

In this section, we review the model of the MASH 1–1–1 DDSM (DDSM3) shown in Fig. 1. The 1-bit quantizer in each constituent EFM1 implements the following operation:

$$y_i[n] = Q(v_i[n]) = \begin{cases} 0, & v_i[n] < M, \\ 1, & v_i[n] \geq M, \end{cases} \tag{1}$$

where M is the step size of the quantizer.

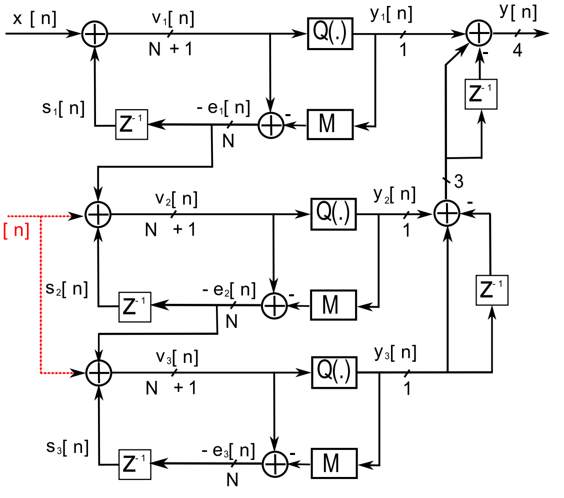

Fig. 3. Solution proposed by Gonzalez-Diaz et al. for "efficiently" dithering a MASH 1–1–1 [12].

3.1. Time domain analysis

The value of the state $s_i[n]$ of the i th EFM1 is given by

$$\begin{aligned} s_i[n] &= v_i[n-1] \bmod M \\ &= (x_i[n-1] + s_i[n-1]) \bmod M, \end{aligned} \tag{2}$$

where mod denotes the modulus operator.

Expanding the above equation with its indices yields

$$\begin{aligned} &s_i[1] = (x_i[0] + s_i[0]) \bmod M \\ &s_i[2] = (x_i[1] + s_i[1]) \bmod M \\ &\vdots \\ &s_i[n-1] = (x_i[n-2] + s_i[n-2]) \bmod M \\ &s_i[n] = (x_i[n-1] + s_i[n-1]) \bmod M. \end{aligned}$$

Substituting the value of $s_i[n-1]$ in $s_i[n]$ and repeating this operation down to index 0 yields

$$s_i[n] = \left(s_i[0] + \sum_{k=0}^{n-1} x_i[n] \right) \bmod M. \tag{3}$$

We will use this expression in Sections 4–6 to determine the underlying periodicity of the DDSM. Recall that a signal s_i is periodic with period L_s if there exists $L_s > 0$ such that $s_i[n + L_s] = s_i[n]$ for all n [7]. Without loss of generality, we will classify as periodic with period L_s the signal for which $s_i[L_s] = s_i[0]$.

Additive least significant bit (LSB) dithering is commonly used to break limit cycles in a DDSM. In this case, a 1-bit dither signal $d[n]$, filtered by a shaping filter, $V(z) = (1 - z^{-1})^R$, where R is a positive integer, is added to the LSBs of the bounded desired signal, $x[n]$. The sum $x_1[n] = x[n] + d[n]$ serves as the input to the entire MASH system. Pamarti et al. have proven that the output spectrum of the DDSM remains spur-free provided that d is random, $l \geq 2$, and $R \leq l-2$, where l is the order of the DDSM [11]. Therefore, we will consider in this paper only zeroth- and first-order shaped dithering of a MASH 1–1–1 DDSM, although the methods we use are sufficiently general to be extendable to higher-order structures.

In practice, the dither sequence is typically generated by a linear feedback shift register (LFSR) which produces a *periodic* pseudorandom signal rather than a truly random signal. The period of this signal is determined by the number of bits used in the LFSR implementation. In this work, we consider a dither sequence generated by a maximum length 8-bit LFSR with a generator polynomial of $p(x)=x^8+x^6+x^5+x^4+1$ and denote the initial condition of the LFSR as $s_d[0]$.[1] In the following, we will assume that the dither signal is *periodic* with period L_{sd}.

3.2. Frequency domain analysis

In the Z-domain, the outputs of the three stages of the MASH 1–1–1 DDSM are given by

$$Y_1(z)=\frac{1}{M}\left(X(z)+V(z)D(z)+\left(1-z^{-1}\right)E_1(z)\right), \tag{4}$$

$$Y_2(z)=\frac{1}{M}\left(-E_1(z)+\left(1-z^{-1}\right)E_2(z)\right), \tag{5}$$

$$Y_3(z)=\frac{1}{M}\left(-E_2(z)+\left(1-z^{-1}\right)E_3(z)\right), \tag{6}$$

where X, D, E_i, and Y_i are the Z-transforms of x, d, e_i, and y, respectively. The output y is given by

$$\begin{aligned} Y(z) &= Y_1(z)+\left(1-z^{-1}\right)Y_2(z)+(1-z^{-1})^2Y_3(z) \\ &= \frac{1}{M}\left((X(z)+V(z)D(z))+(1-z^{-1})^3E_3(z)\right). \end{aligned} \tag{7}$$

Note that the quantization noise $E_3(z)$ is highpass filtered by the noise transfer function $\mathrm{NTF}(z)=(1/M)(1-z^{-1})^3$. By contrast, the filtered dither signal $V(z)D(z)$ experiences the same allpass transfer function as the signal, i.e. $\mathrm{STF}(z)=1/M$. If the period of the dither signal is too low, it will appear as a periodic signal at the output.

Fig. 4(a) shows the simulated spectrum of the output y of an undithered MASH 1–1–1 with $M=2^4$, $I=8$, and $x[n]=I$ for all n. The spectrum is characterized by a small number of tones. Fig. 4(b) shows the simulated spectrum of the same MASH 1–1–1 with $M=2^4$ and $I=8$, this time including first-order shaped additive LSB dither. Here, $L_{sd}=255$ (8-bit LFSR). For illustrative purposes, small values have been chosen for both M and L_{sd}, making both the shaped dither and quantization "noise" signals visibly periodic, with periods L_{sd} and $2M\times L_{sd}$, respectively [16].

We have overlaid in Fig. 4(a) and (b) a solid curve that defines the third-order shaped white (quantization) noise [15]. We have also plotted a dashed curve in Fig. 4(b) which defines the envelope of the LFSR-related spurs, the formula for which we will derive later in this paper.

Note that dithering has improved the situation in (b) relative to (a), in the sense that the quantization noise has been spread over a larger number of tones, resulting in lower power per tone. However, the dithered spectrum is nonetheless characterized by spurs which protrude significantly above the ideal white noise spectral envelope.

[1] In practice, a larger number of bits would be used in the LFSR. We choose 8 bits to exaggerate the resulting effects for illustrative purposes.

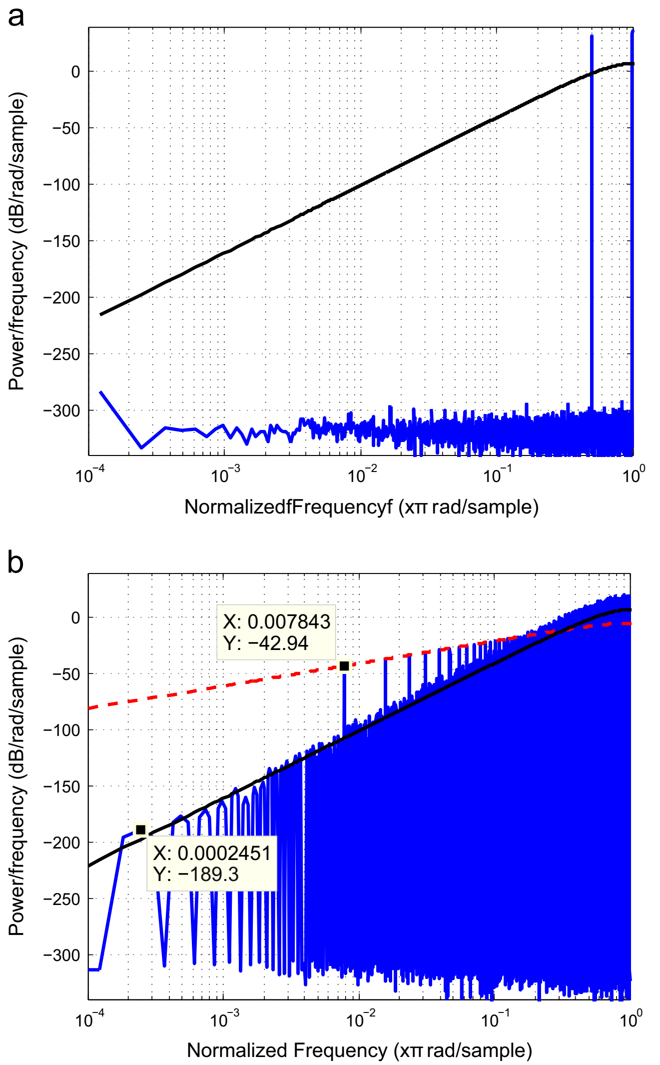

Fig. 4. Output spectra of the MASH 1–1–1 (a) without and (b) with first-order shaped LSB dither. The solid curve is the ideal spectral envelope of third-order shaped white (quantization) noise [16]. The dashed curve indicates the envelope of the LFSR-related spurs. (a) $I = 8$ with no dither; (b) $I = 8$ with first-order shaped additive LSB dither.

The fundamental frequency associated with e_3 in Fig. 4(b) is $f_s/(2M \times L_{sd}) = 2\pi/(2 \times 16 \times 255) = 0.0002451\pi$ rad/sample; this corresponds to the lowest tone in the spectrum. The fundamental associated with the dither signal is $f_s/L_{sd} = 2\pi/255 = 0.007843\pi$ rad/sample.

In the following sections, we will show how to calculate explicitly the period of the output signal in a MASH 1–1–1 DDSM with additive filtered LSB dither and the corresponding envelope of the spurs associated with the dither input.

4. MASH DDSM with additive zeroth-order LSB dither

4.1. Stage 1

The state s_1 of the first stage can be written as

$$s_1[n] = \left(s_1[0] + \sum_{k=0}^{n-1} x_1[n] \right) \mathrm{mod}\ M, \tag{8}$$

where $x_1[n]$ is the input to the first stage. Assuming a constant input to the MASH DDSM, i.e. $x[n]=I$ for all n, and zeroth-order shaped dither (i.e. $R=0$), if the first stage is started from state $s_1[0]$ then, after L_{s1} iterations, we have that

$$s_1[L_{s1}] = \left(s_1[0] + \sum_{k=0}^{L_{s1}-1} (I + d[k]) \right) \mathrm{mod}\ M. \tag{9}$$

s_1 is periodic with period L_{s1} if $s_1[L_{s1}] = s_1[0]$. Equivalently,

$$\left(\sum_{k=0}^{L_{s1}-1} (I + d[k]) \right) \mathrm{mod}\ M = 0. \tag{10}$$

In this case, the condition (10) becomes

$$\left(L_{s1} I + \sum_{k=0}^{L_{s1}-1} d[k] \right) \mathrm{mod}\ M = 0. \tag{11}$$

Let us assume that L_{s1} is an integer multiple of L_{sd}, namely

$$L_{s1} = K_1 L_{sd}, \tag{12}$$

where K_1 is a positive integer. The periodicity condition reduces to

$$\left(K_1 L_{sd} I + K_1 \sum_{k=0}^{L_{sd}-1} d[k] \right) \mathrm{mod}\ M = 0. \tag{13}$$

In the case of a b-bit LFSR that is designed to produce a maximum length sequence, the b-bit state space has 2^b states [17]. If the LFSR is started from the zero state, it stays there indefinitely. If started from any other state, it cycles through the remaining $2^b - 1$ states in a periodic manner; the corresponding dither signal d_b is periodic with period $L_{sd} = 2^b - 1$. Therefore, the expected value $\overline{d_b}$ of the binary dither signal satisfies

$$\overline{d_b} = \frac{2^{b-1}}{2^b - 1}. \tag{14}$$

Thus, the average value of the output of an eight-bit LFSR is 128/255. Furthermore, we have that

$$\sum_{k=0}^{L_{sd}-1} d_b[k] = 2^{b-1}. \tag{15}$$

Substituting Eq. (15) into Eq. (13) yields

$$K_1 (L_{sd} I + 2^{b-1}) \ \mathrm{mod}\ M = 0, \tag{16}$$

where

$$L_{sd} = 2^b - 1. \tag{17}$$

Solving Eq. (16) yields

$$K_1 = \frac{M}{\mathrm{GCD}(L_{sd}I + 2^{b-1}, M)}, \tag{18}$$

where $\mathrm{GCD}(x, y)$ denotes the greatest common divisor of integers x and y. Consequently, the cycle length of the EFM1 with non-shaped dither is given by

$$L_{s1} = \frac{L_{sd}M}{\mathrm{GCD}(L_{sd}I + 2^{b-1}, M)}. \tag{19}$$

We make several observations concerning the first stage with zeroth-order shaped dither [16]:

- The period of s_1 depends explicitly on the input.
- The period is independent of the initial state $s_1[0]$ of the stage.
- The period is independent of the initial state $s_d[0]$ of the dither generator.
- The spectrum of y_1 contains tones due to feedthrough of the dither signal d.

Fig. 5 shows the simulated PSD of y_1 for the case $I=3$ and $M=2^4$. The fundamental components of the first-order shaped quantization noise spectrum of the first stage (associated with L_{s1}) are located at 2/4080 (0.0004902) times the normalized reference frequency, as expected. Note the presence of large spurs at multiples of 2/255 (0.007843) times the normalized reference frequency due to the inherent periodicity of the dither generator. Note also that the contribution from the PRBS generator is not shaped; rather, it is flat.

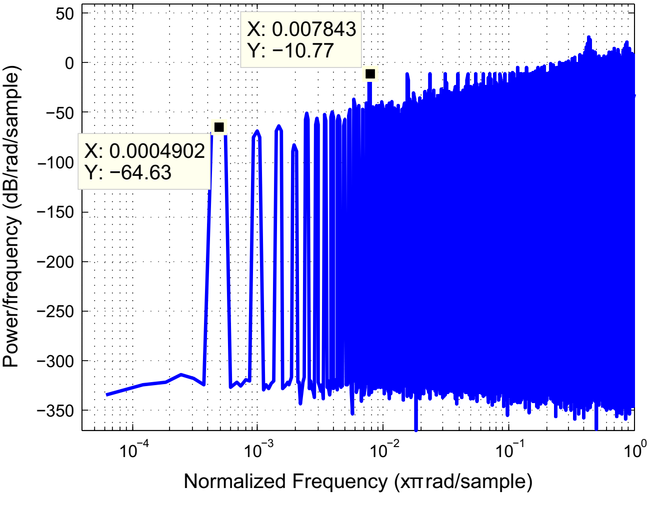

Fig. 5. Simulated PSD at the output of the first stage (y_1) of a MASH 1–1–1 DDSM with $M=2^4$, $I=3$, $s_1[0]=0$, $s_2[0]=0$, $s_3[0]=0$, and $s_d[0]=00000001_2$ in the case of zeroth-order shaped dither at the input.

4.2. Stage 2

If the second stage is started from state $s_2[0]$ then, after L_{s2} iterations, we have that

$$s_2[L_{s2}] = \left(s_2[0] + \sum_{k_2=0}^{L_{s2}-1} s_1[k_2] \right) \bmod M$$
$$= \left(s_2[0] + L_{s2} s_1[0] + \sum_{k_2=0}^{L_{s2}-1} \sum_{k_1=0}^{k_2-1} x_1[k_1] \right) \bmod M. \tag{20}$$

s_2 is periodic with period L_{s2} if $s_2[L_{s2}] = s_2[0]$. Equivalently,

$$\left(L_{s2} s_1[0] + \sum_{k_2=0}^{L_{s2}-1} \sum_{k_1=0}^{k_2-1} x[k_1] \right) \bmod M = 0. \tag{21}$$

When $x_1[k_1] = I + d[k_1]$, the periodicity condition of the second stage reduces to

$$\left(L_{s2} s_1[0] + \frac{L_{s2}(L_{s2}-1)}{2} I + \sum_{k_2=0}^{L_{s2}-1} \sum_{k_1=0}^{k_2-1} d[k_1] \right) \bmod M = 0. \tag{22}$$

With the assumption that $L_{s2} = K_2 L_{s1} = K_2 K_1 L_{sd}$, where K_1 and K_2 are positive integers, we have that

$$\left(K_2 L_{s1} s_1[0] + \frac{K_2(K_2 L_{s1}-1)}{2} L_{s1} I + \sum_{k_2=1}^{K_2 K_1 L_{sd}} \sum_{k_1=1}^{k_2} d[k_1] \right) \bmod M = 0. \tag{23}$$

We make the following observation concerning the period of the output of the second stage:

- The period of s_2 depends explicitly not only on the input, but also on the initial state $s_1[0]$ of the first stage and the initial state $s_d[0]$ of the dither generator.

4.3. Stage 3

If the third stage is started from state $s_3[0]$ then, after L_{s3} iterations, we have that

$$s_3[L_{s3}] = \left(s_3[0] + \sum_{k_3=0}^{L_{s3}-1} s_2[k_3] \right) \bmod M$$
$$= \left(s_3[0] + L_{s3} s_2[0] + \frac{L_{s3}(L_{s3}-1)}{2} s_1[0] + \sum_{k_3=0}^{L_{s3}-1} \sum_{k_2=0}^{k_3-1} \sum_{k_1=1}^{k_2} x[k_1] \right) \bmod M. \tag{24}$$

Imposing the condition $s_3[L_{s3}] = s_3[0]$ yields

$$\left(L_{s3} s_2[0] + \frac{L_{s3}(L_{s3}-1)}{2} s_1[0] + \sum_{k_3=0}^{L_{s3}-1} \sum_{k_2=0}^{k_3-1} \sum_{k_1=0}^{k_2-1} x[k_1] \right) \bmod M = 0. \tag{25}$$

When $x[k_1] = I + d[k_1]$, the periodicity condition of the third stage reduces to

$$\left(L_{s3} s_2[0] + \frac{L_{s3}(L_{s3}-1)}{2} s_1[0] + \frac{L_{s3}(L_{s3}-1)(L_{s3}-2)}{6} I + \sum_{k_3=0}^{L_{s3}-1} \sum_{k_2=0}^{k_3-1} \sum_{k_1=0}^{k_2-1} d[k_1] \right) \bmod M = 0. \tag{26}$$

We make the following observation concerning the period of the output of the third stage:

- The period of s_3 depends explicitly on the input, the initial states $s_1[0]$ and $s_2[0]$ of the first and second stages, and the initial state $s_d[0]$ of the dither generator.

In the worst case, the output period can be as small as $4 \times L_{sd}$; in the best case, it is $2M \times L_{sd}$.

4.4. Output

The output y is formed by combining the outputs y_1, y_2, and y_3 of the three EFM1 stages in the error cancelation network. Fig. 6 shows the simulated PSD at the output y of the DDSM with $I=9$, zero initial state, and with the LFSR's initial condition set to 00000001. The output of the undithered MASH has the maximum period of $2M$ in this case [15].

The fundamental component of the first-order shaped quantization noise in the third stage (associated with L_{s3}) is located at 2/8160 times the normalized reference frequency, as expected. Note the presence of large spurs at multiples of 2/255 (0.007843) times the normalized reference frequency due to the inherent periodicity of the dither generator. These were already present in y_1 (see Fig. 5) and have not been eliminated by the MASH structure. The solid and dashed curves represent the ideal spectrum of the third-order shaped white quantization noise and the zeroth-order shaped dither signal, respectively.

Note that the powers of the spurs due to the periodicity of the dither signal are significantly above the white noise approximation. Thus, although the dithering has extended the period of the

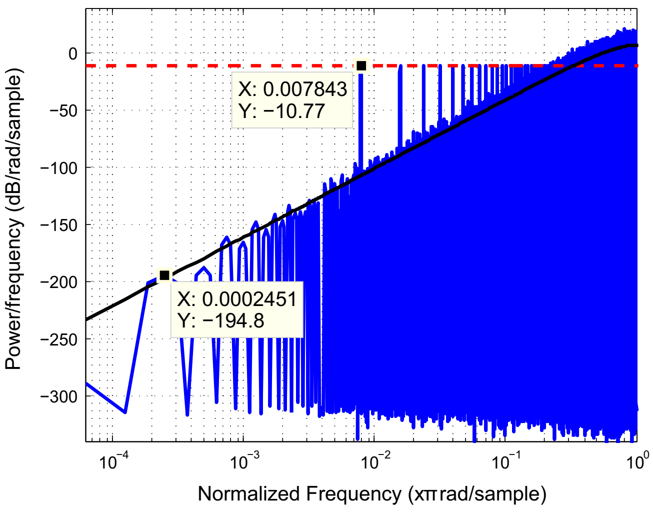

Fig. 6. Simulated PSD at the output (y) of a MASH 1–1–1 DDSM with $M = 2^4$, $I=9$, $s_1[0] = 0$, $s_2[0] = 0$, $s_3[0] = 0$, and $s_d[0] = 00000001_2$ in the case of zeroth-order shaped dither at the input.

output signal from $2M \times L_{sd}$, the unfiltered dither signal itself has been passed directly to the output of the DDSM.

5. Power per tone in the dithered signal

The dashed curve in Fig. 6 can be derived using Parseval's theorem. Parseval's theorem states that the energy or power in the time-domain representation of a periodic signal with fundamental period L_s is equal to the energy or power in the frequency-domain representation, i.e.,

$$\frac{1}{L_s}\sum_{n=1}^{L_s}|x[n]|^2=\sum_{k=1}^{L_s}|X[k]|^2, \tag{27}$$

where $x[n]$ is the discrete time series and $X[k]$ is its corresponding discrete-time Fourier series (DTFS).

In the case of the zeroth-order dithered MASH 1–1–1 with a maximum length PRBS generator, the length of the dither sequence is $L_{sd}=2^b-1$. Thus,

$$\frac{1}{L_{sd}}\sum_{n=1}^{L_{sd}}|d_e[n]|^2=\sum_{k=1}^{L_{sd}}|D_e[k]|^2, \tag{28}$$

where

$$d[n]\in 0,1 \tag{29}$$

and

$$d_e[n]=d[n]-\overline{d_b} \tag{30}$$

is the DC-free signal. Recall that

$$\overline{d_b}=\frac{2^{b-1}}{2^b-1}. \tag{31}$$

The total energy in one cycle of the DC-free dither signal is given by

$$\begin{aligned}\sum_{n=1}^{L_{sd}}|d_e[n]|^2 &= \sum_{n=1}^{L_{sd}}|d[n]-\overline{d_b}|^2\\ &= |d[1]-\overline{d_b}|^2+|d[2]-\overline{d_b}|^2+\cdots+|d[L_{sd}]-\overline{d_b}|^2\\ &= \underbrace{(2^{b-1}-1)}_{\text{No. of 0 s in one period}}\times\overline{d_b}^2\\ &\quad+\underbrace{2^{b-1}}_{\text{No. of 1 s in one period}}\times(1-2\overline{d_b}+\overline{d_b}^2)\\ &= L_{sd}\overline{d_b}^2-2^b\overline{d_b}+2^{b-1}.\end{aligned} \tag{32}$$

There are L_{sd} bins in the frequency range $[0,2\pi]$ in an $L_{sd}-$point FFT. $L_{sd}-1$ of these represent the signal power spectrum. We assume that the spectrum of the PRBS is white; therefore, the power of each bin is given by

$$S=\frac{1}{(L_{sd}-1)L_{sd}}\sum_{n=1}^{L_{sd}}|d_e[n]|^2$$

$$= \frac{L_{sd}\overline{d_b}^2 - 2^b\overline{d_b} + 2^{b-1}}{L_{sd}(L_{sd}-1)}. \tag{33}$$

The processing gain PG of the window w [18] is defined as the ratio of the output signal-to-noise ratio to the input signal-to-noise ratio, and is given by

$$PG = \frac{S_o/N_o}{S_i/N_i}$$
$$= \frac{(\sum_n w[n])^2}{\sum_n w^2[n]}, \tag{34}$$

where $w[n]$ is the window sequence. In particular, the processing gain of the Hanning window is

$$PG = \left(\frac{1}{1.23}\right)^2. \tag{35}$$

Thus, the power per tone in the FFT is

$$S_{PSD} = \left(\frac{L_{sd}\overline{d_b}^2 - 2^b\overline{d_b} + 2^{b-1}}{L_{sd}\cdot(L_{sd}-1)}\right) N_f\ PG, \tag{36}$$

where N_f is the size of the FFT used to process the data.

The dither signal passes through the signal transfer function $\mathrm{STF}(z) = 1/M$ before appearing at the output. Hence, the dashed curve Fig. 6 is defined by

$$S_0 = \left(\frac{L_{sd}\overline{d_b}^2 - 2^b\overline{d_b} + 2^{b-1}}{L_{sd}(L_{sd}-1)}\right)\left(\frac{N_f\ PG}{M^2}\right). \tag{37}$$

As L_{sd} is increased, the simulated spectrum more closely matches the ideal spectrum defined by Eq. (7). This is illustrated in Fig. 7 which shows the simulated output PSD with the same set of parameters as before, but this time using a 20-bit maximum length LFSR instead of an 8-bit one.

6. MASH DDSM with additive first-order LSB dither

In the case of a MASH 1–1–1 DDSM, first-order shaping of the additive dither signal can be achieved in two ways. The scheme of Fig. 1 could be used with $V(z) = 1 - z^{-1}$. Alternatively, a zeroth-order shaped dither signal can be added to the input of the second stage, as shown in Fig. 8; this produces equivalent first-order shaped dither without requiring an explicit filter $V(z)$. In both cases, the output is given by

$$Y(z) = \frac{1}{M}\left(X(z) + \left(1 - z^{-1}\right)D(z) + (1 - z^{-1})^3 E_3(z)\right). \tag{38}$$

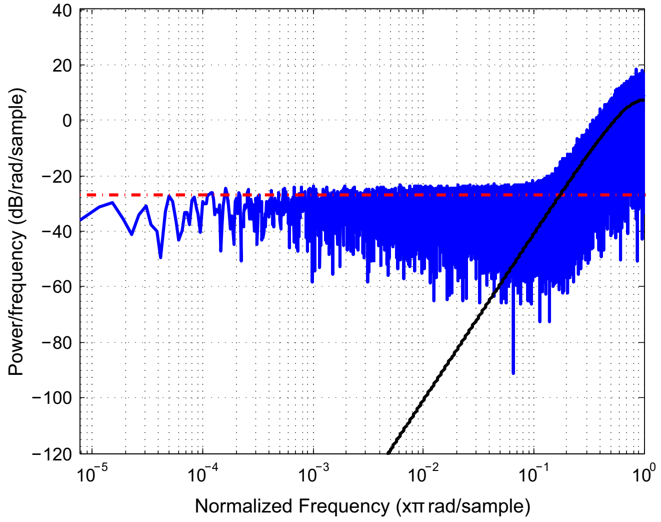

Fig. 7. Simulated PSD at the output (y) of a MASH 1–1–1 DDSM with $M=2^4$, $X=9$, $s_1[0]=s_2[0]=s_3[0]=0$, and $s_d[0]=00000000000000000001_2$ in the case of zeroth-order shaped dither at the input. A 20-bit LFSR has been used in this example.

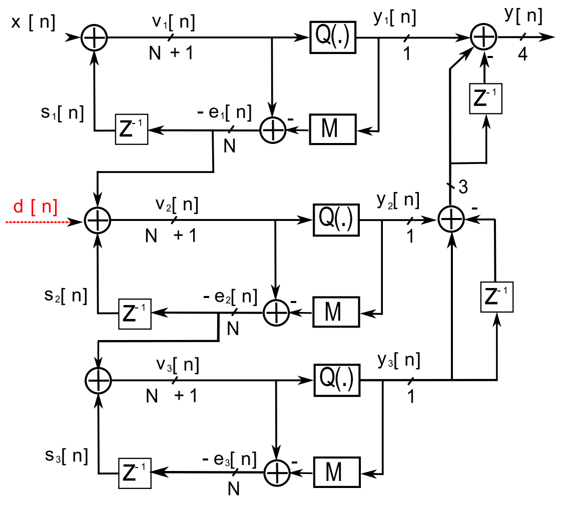

Fig. 8. Block diagram of a MASH 1–1–1 DDSM (DDSM3) with first-order shaped additive LSB dither.

6.1. Stage 1

Assuming a constant input to the MASH DDSM in Fig. 8 of $x[n]=I$, if the first stage is started from state $s_1[0]$ then, after L_{s1} iterations, we have that

$$s_1[L_{s1}]=\left(s_1[0]+\sum_{k=0}^{L_{s1}-1} L\right) \bmod M. \tag{39}$$

Imposing the condition $s_1[L_{s1}] = s_1[0]$ yields

$$(L_{s1}I) \bmod M = 0. \tag{40}$$

Solving the above equation yields

$$L_{s1} = \frac{M}{\mathrm{GCD}(I, M)}. \tag{41}$$

This undithered EFM1 has been studied extensively in the literature [15]. When M is a power of two, the maximum cycle length (M) is achieved when I is odd. The minimum cycle length (2) occurs when $I = M/2$ (excluding the trivial $I=0$ case).

6.2. Stage 2

If the second stage is started from state $s_2[0]$ then, after L_{s2} iterations, we have that

$$s_2[L_{s2}] = \left(s_2[0] + \sum_{k_2=0}^{L_{s2}-1} (s_1[k_2] + d[k_2])\right) \bmod M = \left(s_2[0] + L_{s2}s_1[0] + \frac{L_{s2}(L_{s2}-1)}{2}I + \sum_{k_2=0}^{L_{s2}-1} d[k_2]\right) \bmod M. \tag{42}$$

s_2 is periodic with period L_{s2} if $s_2[L_{s2}] = s_2[0]$. Equivalently,

$$\left(L_{s2}s_1[0] + \frac{L_{s2}(L_{s2}-1)}{2}I + \sum_{k_2=0}^{L_{s2}-1} d[k_2]\right) \bmod M = 0. \tag{43}$$

With the assumption that $L_{s2} = K_2 L_{s1} L_{sd}$, the periodicity condition of the second stage reduces to

$$\left(K_2 L_{s1} L_{sd} s_1[0] + \frac{K_2(K_2 L_{s1} L_{sd} - 1)}{2} L_{s1} L_{sd} I + K_2 L_{s1} 2^{b-1}\right) \bmod M = 0. \tag{44}$$

The second term in Eq. (44) can be simplified in the two cases where L_{s1} is even or odd [15]. In the case of odd L_{s1}, we have

$$L_{s2} = \left(\frac{L_{sd}M}{\mathrm{GCD}(M, (L_{sd}s_1[0] + 2^{b-1})L_{s1})}\right)\left(\frac{M}{\mathrm{GCD}(I, M)}\right). \tag{45}$$

When L_{s1} is even,

$$L_{s2} = \left(\frac{L_{sd}2M}{\mathrm{GCD}(2M, ((2s_1[0] - I)L_{sd} + 2^b)L_{s1})}\right)\left(\frac{M}{\mathrm{GCD}(I, M)}\right). \tag{46}$$

6.3. Stage 3

If the third stage is started from state $s_3[0]$ then, after L_{s3} iterations, we have that

$$\begin{aligned} s_3[L_{s3}] &= \left(s_3[0] + \sum_{k_3=0}^{L_{s3}-1} s_2[k_3]\right) \bmod M \\ &= \left(s_3[0] + L_{s3}s_2[0] + \frac{L_{s3}(L_{s3}-1)}{2}s_1[0] + \sum_{k_3=0}^{L_{s3}-1}\sum_{k_2=0}^{k_3-1}\sum_{k_1=0}^{k_2-1} x[k_1] + \sum_{k_3=0}^{L_{s3}-1}\sum_{k_2=0}^{k_3-1} d[k_2]\right) \bmod M. \end{aligned} \tag{47}$$

s_3 is periodic with period L_{s3} if $s_3[L_{s3}] = s_3[0]$. Consequently, for a constant input I, we obtain

$$\left(L_{s3}s_2[0] + \frac{L_{s3}(L_{s3}-1)}{2}s_1[0] + \frac{L_{s3}(L_{s3}-1)(L_{s3}-2)}{6}I + \sum_{k_3=0}^{L_{s3}-1}\sum_{k_2=0}^{k_3-1} d[k_2] \right) \bmod M = 0. \tag{48}$$

As in the case of additive zeroth-order LSB dither, the period of s_3 depends explicitly on the input, the initial states $s_1[0]$ and $s_2[0]$ of the first and second stages, and the initial state $s_d[0]$ of the dither generator. In the worst case, the DDSM's output period with first-order additive LSB dither can be as small as $4 \times L_{sd}$ in the worst case; in the best case, it is $2M \times L_{sd}$.

6.4. Output

Fig. 9 shows the simulated PSD at the output of a MASH 1–1–1 DDSM with $M = 2^4$, $I=7$, $s_1[0] = 2$, $s_2[0] = 1$, and $s_3[0] = 0$ in the case of zeroth-order shaped dither added into the second stage (equivalently, first-order shaped dither added to the input). An 8-bit LFSR ($b=8$) initialized to the value 00000001 is used to generate the dither sequence. The fundamental component of the shaped quantization noise spectrum (associated with L_{s3}) is located at 2/8160 (0.0002451) times the normalized reference frequency, as expected. Large spurs due to the inherent periodicity of the dither generator appear at multiples of 2/255 (0.007843) times the normalized reference frequency. Note also that this contribution from the PRBS is first-order shaped. Indeed, the dashed curve in Fig. 9 is defined by

$$S_1(f) = S_0\left(2\sin\left(\frac{\pi f}{f_s}\right)\right)^2, \tag{49}$$

where S_0 is given by Eq. (37).

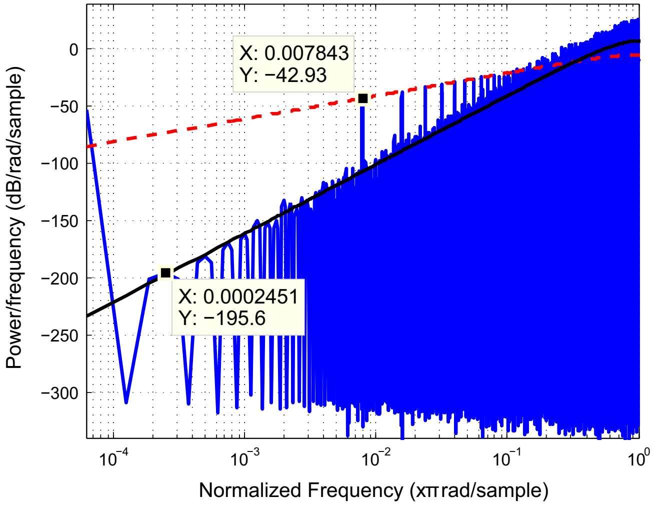

Fig. 9. Simulated PSD at the output of a MASH 1–1–1 DDSM with $M = 2^4$, $I=7$, $s_1[0] = 2$, $s_2[0] = 1$, $s_3[0] = 0$, and $s_d[0] = 00000001_2$ in the case of zeroth-order shaped dither at the input of the second stage.

7. Discussion

Note that no dither tones will be visible in the output spectrum of the MASH 1–1–1 DDSM if

$$S_1(f) < \frac{1}{12}\left(2\sin\left(\frac{\pi f}{f_s}\right)\right)^6 \tag{50}$$

at $f = f_s/L_{sd}$.

For sufficiently large dither period L_{sd}, $\sin(\pi/L_{sd}) \approx \pi/L_{sd}$ and Eq. (50) simplifies to

$$\left(\frac{L_{sd}\overline{d_b}^2 - 2^b\overline{d_b} + 2^{b-1}}{L_{sd}(L_{sd}-1)}\right)\left(\frac{N_f PG}{M^2}\right)\left(\frac{2\pi}{L_{sd}}\right)^2 < \frac{1}{12}\left(\frac{2\pi}{L_{sd}}\right)^6. \tag{51}$$

Furthermore, $\overline{d_b} \approx 2^{-1}$ and $L_{sd} \approx (L_{sd}-1) \approx 2^b$, giving

$$\left(\frac{1}{4L_{sd}}\right)\left(\frac{N_f PG}{M^2}\right) < \frac{1}{12}\left(\frac{2\pi}{L_{sd}}\right)^4. \tag{52}$$

In the special case when $N_f = 2M \times L_{sd}$, we have an approximate upper bound on L_{sd}:

$$L_{sd} < 2\pi\sqrt[4]{\frac{M}{6PG}}. \tag{53}$$

Fig. 10 shows output spectra (in blue) in two extreme cases: (a) when L_{sd} is sufficiently small that the inequality (53) is satisfied, and (b) when L_{sd} is very large. In the first case, the dither tones (shown in red) lie below the solid black curve corresponding to the third-order shaped quantization noise of the DDSM. In the second case, the dither tones, and consequently the output, follow a first-order shaped envelope (shown dashed in green) at low frequencies. Unlike Fig. 9, no individual spurs are visible in the output spectrum in either case.

8. Conclusion

A pseudorandom dither generator based on an LFSR produces a periodic output. When this signal is used to dither a DDSM with a constant input, the overall system has a periodic output. The harmonic components of the output, and the so-called spurious tones, are influenced by the length of the PRB sequence, the initial condition of the LFSR, and the input, initial conditions and modulus of the DDSM. In the worst case, the output is periodic with a period of $4 \times L_{sd}$, where L_{sd} is the length of the PRB sequence. Furthermore, spurs appear in the output spectrum at f_s/L_{sd} and its harmonics.

For the first stage, when dither is applied to the input of the MASH DDSM3, the best result is that the cycle length is increased, thereby reducing the heights of most spurs. However, when the dither is unfiltered, there is an increased noise floor; in addition, strong spurs still remain due to the inherent periodicity of the pseudorandom dither signal. When the dither is filtered, the effects resulting from the periodic nature of the dither signal are reduced at low frequencies. Even with filtering, however, spurs that result from the underlying undithered structure still remain at the output of the first stage.

In the second error feedback modulator, the cycle lengths are generally longer than those of the first stage. The evolution of the second stage is similar to that of the first stage in terms of the effects of applied dither, but the initial state of the first stage and the initial values of the LFSR can have a significant effect on the cycle length of the second stage.

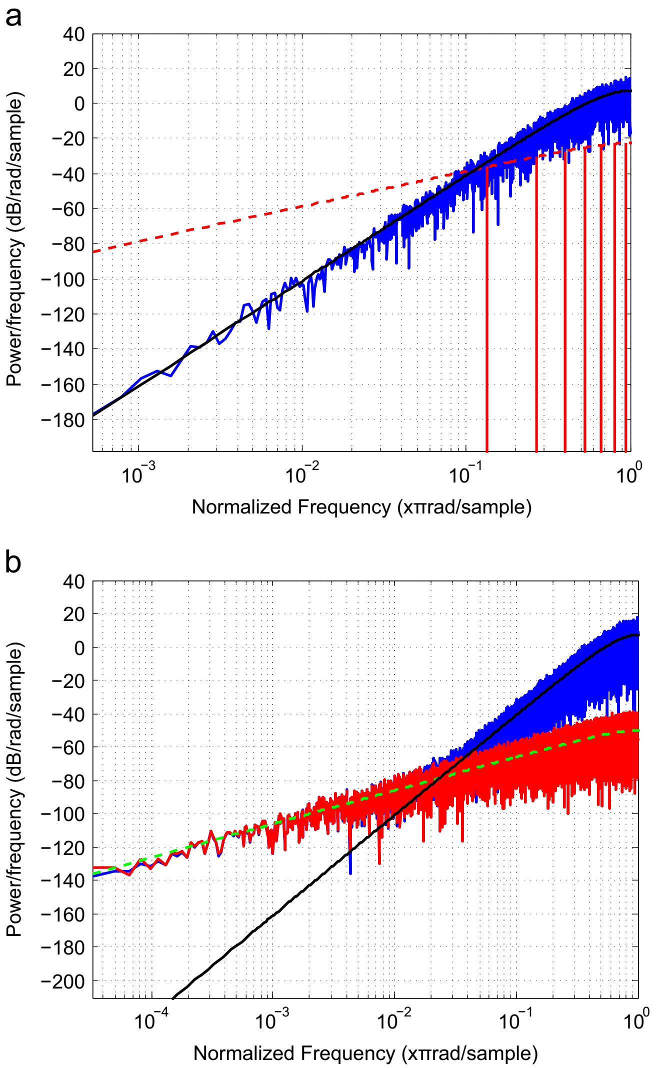

Fig. 10. Simulated PSDs (shown in blue) at the output of a MASH 1–1–1 DDSM with $M = 2^8$, $I = M/2$, $s_1[0] = 1$, $s_2[0] = 0$, $s_3[0] = 0$ in the case of zeroth-order shaped dither at the input of the second stage. (a) 4-bit LFSR with $s_d[0] = 1100_2$ and $N_f = 2M \times L_{sd}$; the (few) dither tones are shown in red, with a dashed envelope defined by Eq. (53). (b) 20-bit LFSR with $s_d[0] = 11000000000000000000_2$ and $N_f = 32M \times L_{sd}$; the very large number of dither tones leads to an almost continuous curve which is approximated by first-order shaped white noise (shown dashed in green). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

The cycle length of the third and the final stage is critical as the quantization noise from the final stage appears in the output. In the worst case, the state variable s_3 is periodic with a period of $4 \times L_{sd}$. In the best case, s_3 is periodic with a period of $2M \times L_{sd}$. The results are summarized in Table 1. First-order shaped dither added to the input of the first stage and zeroth-order dither added to the input of the second stage yield identical cycle length and output sequence values.

Table 1
Minimum and maximum cycle lengths L_{s3} for MASH DDSM with power-of-2 quantizer modulus M and dither period L_{sd}.

MASH order l	Minimum cycle length	Maximum cycle length
3	$4 \times L_{sd}$	$2M \times L_{sd}$

Consider once again the spectra shown in Fig. 2 which have been produced by the dithered MASH 1–1–1 DDSM in Fig. 3. In this example, $f_s = 25$ MHz, $M = 256$ and $I = 128$. With no dither, the DDSM produces a periodic output with period 4; this results in a fundamental at 6.25 MHz.

With an additive pseudorandom dither signal having a very large value of L_{sd} (the "random signal dither" case), the simulated spectrum approaches the envelope bounded by the dashed green and solid black curves in Fig. 10(b). If L_{sd} is sufficiently small, the dither tones are completely masked by the third-order shaped quantization noise of the DDSM, as shown in Fig. 10(a). For values of L_{sd} between these extremes, tones due to dither feedthrough will be visible in the output spectrum, as shown in Fig. 9.

In summary, although dithering is effective in increasing the cycle lengths in all stages of a MASH DDSM, the periodic nature of a pseudorandom dither generator may manifest itself in the output spectrum if the number of points N_f in the FFT is sufficiently large (equivalently, if the resolution bandwidth of the measured spectrum is sufficiently small).

Acknowledgements

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